

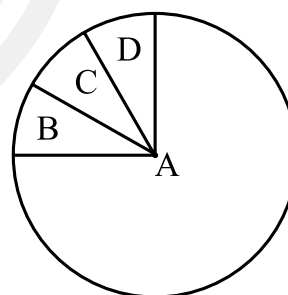
CHAPTER 13

Biodiversity and Conservation

Biodiversity & Patterns of Biodiversity

- Tropical regions show greatest level of species richness because (2024)
 - Tropical latitudes have remained relatively undisturbed for millions of years, hence more time was available for species diversification.
 - Tropical environments are more seasonal.
 - More solar energy is available in tropics.
 - Constant environments promote niche specialization.
 - Tropical environments are constant and predictable.
 Choose the correct answer from the options given below:
 - A, B and E only
 - A, B and D only
 - A, C, D and E only
 - A and B only
- In 'rivet popper hypothesis', Paul Ehrlich compared the rivets in an airplane to: (2023-Manipur)
 - species within a genus
 - genetic diversity
 - ecosystem
 - genera within a family
- Give the correct descending order of organisms with reference to their estimated number found in Amazon forest. (2022 Re)
 - Plants
 - Invertebrates
 - Fishes
 - Mammals
 - Birds
 Choose the correct answer from the options given below
 - (b) > (a) > (c) > (e) > (d)
 - (a) > (b) > (e) > (d) > (c)
 - (a) > (c) > (d) > (b) > (e)
 - (b) > (a) > (e) > (d) > (c)
- Frugivorous birds are found in large numbers in tropical forests mainly because of: (2022 Re)
 - temperature conducive for their breeding
 - lack of niche specialisation
 - higher annual rainfall
 - availability of fruits throughout the year
- Which of the following regions of the globe exhibits highest species diversity? (2020)
 - Madagascar
 - Himalayas
 - Amazon forests
 - Western ghats of India
- According to Robert May, the global species diversity is about: (2020)
 - 20 million
 - 50 million
 - 7 million
 - 1.5 million

- According to Alexander von Humboldt: (2020-Covid)
 - Species richness increases with increasing area, but only up to limit
 - There is no relationship between species richness and area explored
 - Species richness goes on increasing with increasing area of exploration
 - Species richness decreases with increasing area of exploration
- Alexander von Humboldt described for the first time: (2017-Delhi)
 - Ecological Biodiversity
 - Laws of limiting factor
 - Species area relationships
 - Population Growth equation
- Given below is the representation of the extent of global diversity of invertebrates. What groups the four portions (A-D) represent respectively? (2014)



	A	B	C	D
a.	Insects	Molluscs	Crustaceans	Other animal groups
b.	Insects	Crustaceans	Other animal groups	Molluscs
c.	Crustaceans	Insects	Molluscs	Other animal groups
d.	Molluscs	Other animal groups	Crustaceans	Insects

- Which of the following represent maximum number of species among global biodiversity? (2013)
 - Mosses and Ferns
 - Algae
 - Lichens
 - Fungi

Importance & Loss of Biodiversity

11. These are regarded as major causes of biodiversity loss: (2024)

A. Over exploitation
B. Co-extinction
C. Mutation
D. Habitat loss and fragmentation
E. Migration

Choose the correct option:

- a. A, B and E only b. A, B and D only
c. A, C and D only d. A, B, C and D only
12. List of endangered species was released by- (2024)
a. FOAM b. IUCN c. GE AC d. WWF
13. Match List-I with List-II. (2024)

List-I		List-II	
(A)	Robert May	(I)	Species-Area relationship
(B)	Alexander von Humboldt	(II)	Long term ecosystem experiment using out door plots
(C)	Paul Ehrlich	(III)	Global species diversity at about 7 million
(D)	David Tilman	(IV)	Rivet popper hypothesis

Choose the correct answer from the options given below:

- a. A-(I), B-(III), C-(II), D-(IV)
b. A-(III), B-(IV), C-(II), D-(I)
c. A-(II), B-(III), C-(I), D-(IV)
d. A-(III), B-(I), C-(IV), D-(II)
14. Among 'The Evil Quartet', which one is considered the most important cause driving extinction of species? (2023)
a. Co-extinctions
b. Habitat loss and fragmentation
c. Over exploitation for economic gain
d. Alien species invasions
15. Which of the following is/are cause(s) of biodiversity losses? (2023-Manipur)
a. Over-exploitation, habitat loss and fragmentation
b. Climate change only
c. Over-exploitation only
d. Habitat loss and fragmentation only
16. Habitat loss and fragmentation, over exploitation, alien species invasion and co-extinction are causes for: (2022)
a. Natalty b. Population explosion
c. Competition d. Biodiversity loss
17. Which of the following is the most important cause for animals and plants being driven to extinction? (2019)
a. Habitat loss and fragmentation
b. Drought and floods
c. Economic exploitation
d. Alien species invasion

18. Red list contains data or information on: (2016 - II)
a. Threatened species
b. Marine vertebrates only
c. All economically important plants
d. Plants whose products are in international trade
19. Which of the following is correctly matched? (2016 - II)
a. *Parthenium hysterophorus* – Threat to biodiversity
b. Stratification – Population
c. Aerenchyma – *Opuntia*
d. Age pyramid – Biome
20. Which of the following is the most important cause of animals and plants being driven to extinction? (2016 - I)
a. Over-exploitation
b. Alien species invasion
c. Habitat loss and fragmentation
d. Co-extinctions
21. The organisation which publishes the Red List of species is: (2014)
a. WWF b. ICFRE c. IUCN d. UNEP
22. A species facing extremely high risk of extinction in the immediate future is called: (2014)
a. Extinct b. Vulnerable
c. Endemic d. Critically Endangered

Biodiversity Conservation

23. The regions with high level of species richness, high degree of endemism and a loss of 70% of the species and habitat are identified as: (2024 Re)
a. Natural Reserves b. Sacred Groves
c. Biodiversity Hotspots d. Biogeographical Regions
24. Which one of the following is **not** included under *in-situ* conservation? (2024 Re)
a. Wild-life sanctuary b. Botanical garden
c. Biosphere reserve d. National park
25. Cryopreservation technique is used for: (2024 Re)
a. Protection of environment.
b. Protection of Biodiversity hotspots.
c. Preservation of gametes in viable and fertile condition for a long period.
d. *In-situ* conservation.
26. Match List-I with List-II: (2024 Re)

List-I		List-II	
(A)	Biodiversity hotspot	(I)	Khasi and Jantia hills in Meghalaya
(B)	Sacred groves	(II)	World Summit on Sustainable Development 2002
(C)	Johannesburg South Africa	(III)	<i>Parthenium</i>
(D)	Alien species invasion	(IV)	Western Ghats

Choose the **correct** answer from the options given below:

- a. A-IV, B-I, C-II, D-III b. A-II, B-III, C-IV, D-I
c. A-I, B-IV, C-III, D-II d. A-III, B-I, C-II, D-IV

27. The type of conservation in which the threatened species are taken out from their natural habitat and placed in special setting where they can be protected and given special care is called; (2024)
- Semi-conservative method
 - Sustainable development
 - in-situ* conservation
 - Biodiversity conservation
28. The historic Convention on Biological Diversity, 'The Earth Summit' was held in Rio de Janeiro in the year: (2023)
- 2002
 - 1992
 - 1985
 - 1986
29. The World Summit on sustainable development held in 2002 in Johannesburg, South Africa pledged for: (2022 Re)
- Collection and preservation of seeds of different genetic strains of commercially important plants.
 - A significant reduction in the current rate of biodiversity loss.
 - Declaration of more biodiversity hotspots.
 - Increase in agricultural production.
30. Match List-I with List-II: (2022 Re)

List-I		List-II	
(A)	Sacred groves	(I)	Alien species
(B)	Zoological park	(II)	Release of large quantity of oxygen
(C)	Nile perch	(III)	<i>Ex-situ</i> conservation
(D)	Amazon forest	(IV)	Khasi Hills in Meghalaya

Choose the correct answer from the options given below:

- A-IV, B-III, C-II, D-I
 - A-IV, B-III, C-I, D-II
 - A-II, B-IV, C-I, D-III
 - A-IV, B-I, C-II, D-III
31. Western Ghats have a large number of plants and animal species that are not found anywhere else. Which of the following term is used to notify such species? (2022 Re)
- Vulnerable species
 - Threatened species
 - Keystone species
 - Endemic species
32. Which of the following is **not** a method of *ex situ* conservation? (2022)
- Cryopreservation
 - In vitro* fertilization
 - National Parks
 - Micropropagation
33. *In-situ* conservation refers to: (2022)
- Conserve only extinct species
 - Protect and conserve the whole ecosystem
 - Conserve only high risk species
 - Conserve only endangered species
34. In the following in each set a conservation approach and an example of method of conservation are given
- In situ* conservation - Biosphere Reserve
 - Ex situ* conservation - Sacred groves
 - In situ* conservation - Seed bank
 - Ex situ* conservation - Cryopreservation
- Select the option with correct match of approach and method: (2020-Covid)
- A and D
 - B and D
 - A and B
 - A and C

35. Which one of the following is not a method of *in situ* conservation of biodiversity? (2019)
- Biosphere Reserve
 - Wildlife Sanctuary
 - Botanical Garden
 - Sacred Grove
36. The Earth Summit held in Rio de Janeiro in 1992 was called (2019)
- To reduce CO₂ emissions and global warming
 - For conservation of biodiversity and sustainable utilization of its benefits
 - To assess threat posed to native species by invasive weed species
 - For immediate steps to discontinue use of CFCs that were damaging the ozone layer
37. All of the following are included in '*ex-situ* conservation' except: (2018)
- Wildlife safari parks
 - Sacred groves
 - Botanical gardens
 - Seed banks
38. Which one of the following is related to *ex-situ* conservation of threatened animals and plants? (2017-Delhi)
- Wildlife Safari parks
 - Biodiversity hot spots
 - Amazon rainforest
 - Himalayan region
39. The region of Biosphere Reserve which is legally protected and where no human activity is allowed is known as: (2017-Delhi)
- Core zone
 - Buffer zone
 - Transition zone
 - Restoration zone
40. How many hot spots of biodiversity in the world have been identified till date by Norman Myers? (2016 - II)
- 34
 - 43
 - 17
 - 25
41. Cryopreservation of gametes of threatened species in viable and fertile condition can be referred to as: (2015)
- In situ* conservation by sacred groves
 - In situ* cryo-conservation of biodiversity
 - In situ* conservation of biodiversity
 - Advanced *ex-situ* conservation of biodiversity
42. In which of the following both pairs have correct combination? (2015)
- In situ* conservation: Seed Bank
Ex situ conservation: National Park
 - In situ* conservation: Tissue culture
Ex situ conservation: Sacred groves
 - In situ* conservation: National Park
Ex situ conservation: Botanical Garden
 - In situ* conservation: Cryopreservation
Ex situ conservation: Wildlife Sanctuary
43. The species confined to a particular region and not found elsewhere is termed as: (2015 Re)
- Alien
 - Endemic
 - Rare
 - Keystone
44. An example of *ex situ* conservation is: (2014)
- Sacred Grove
 - National Park
 - Seed Bank
 - Wildlife Sanctuary
45. Which one of the following is not used for *ex-situ* plant conservation? (2013)
- Botanical Gardens
 - Field gene banks
 - Seed banks
 - Shifting cultivation

Answer Key

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (c) | 2. (a) | 3. (a) | 4. (d) | 5. (c) | 6. (c) | 7. (a) | 8. (c) | 9. (a) | 10. (d) |
| 11. (b) | 12. (b) | 13. (d) | 14. (b) | 15. (a) | 16. (d) | 17. (a) | 18. (a) | 19. (a) | 20. (c) |
| 21. (c) | 22. (d) | 23. (c) | 24. (b) | 25. (c) | 26. (a) | 27. (d) | 28. (b) | 29. (b) | 30. (b) |
| 31. (d) | 32. (c) | 33. (b) | 34. (a) | 35. (c) | 36. (b) | 37. (b) | 38. (a) | 39. (a) | 40. (a) |
| 41. (d) | 42. (c) | 43. (b) | 44. (c) | 45. (d) | | | | | |

